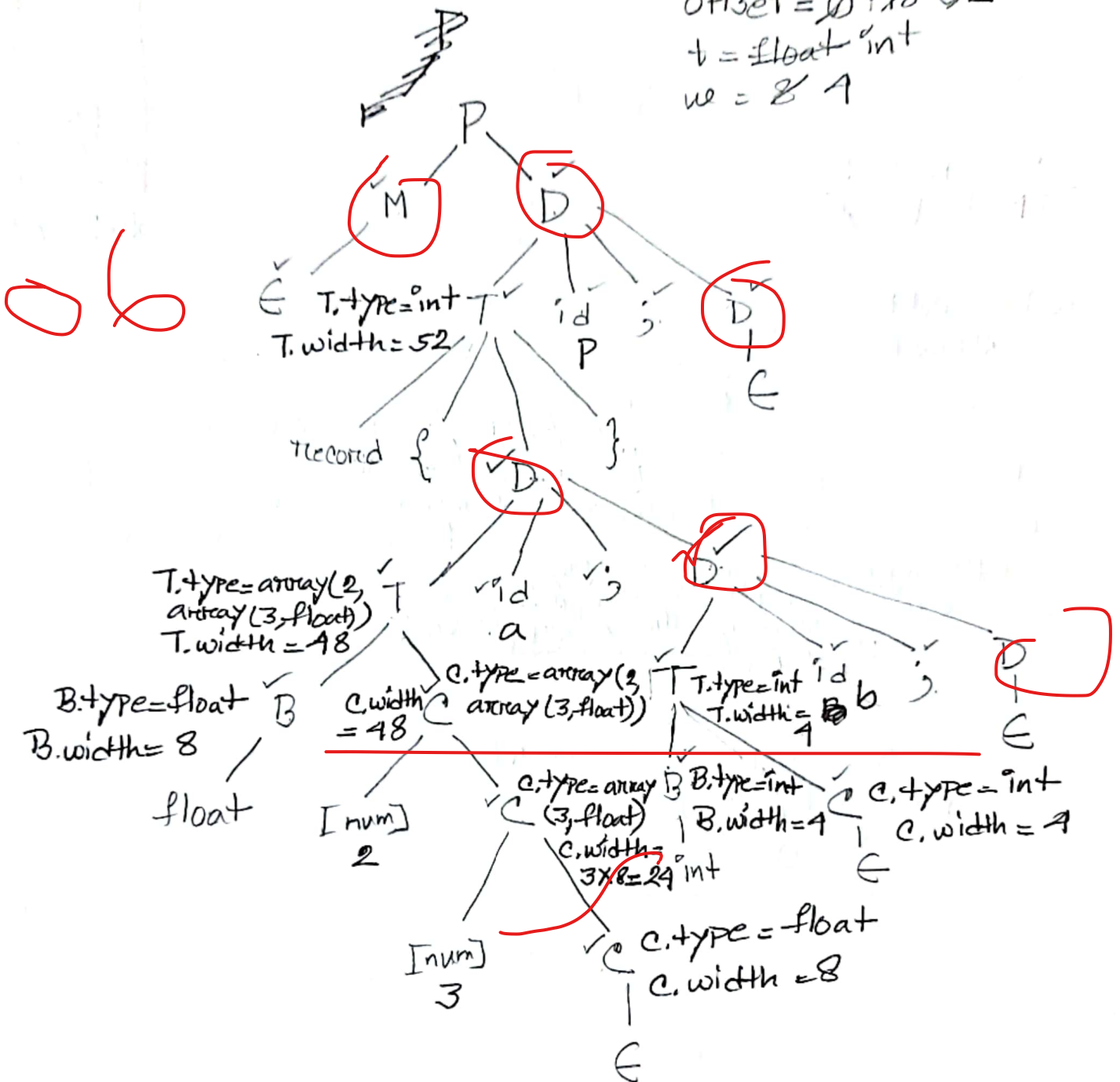


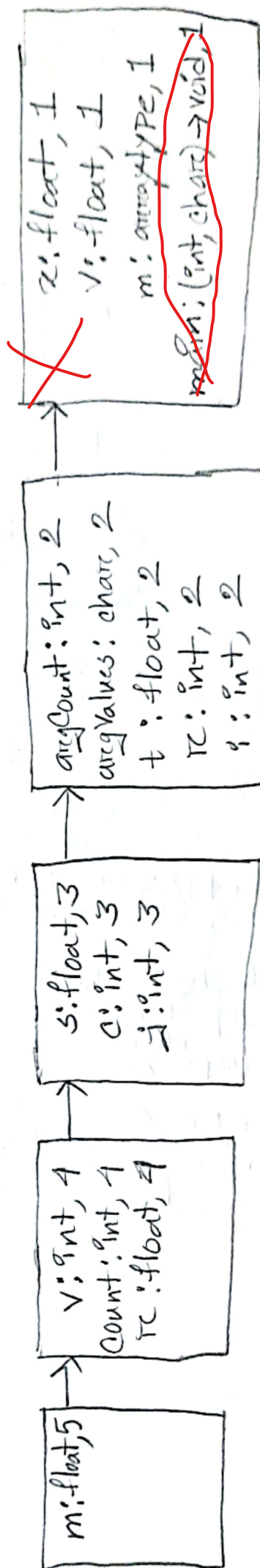
Ans to the Q: No. 1

Input : record { float [2] [3] a; int b; } P;

offset = 0 18 52 104
~~t = float~~ int
~~w = 8~~ 4



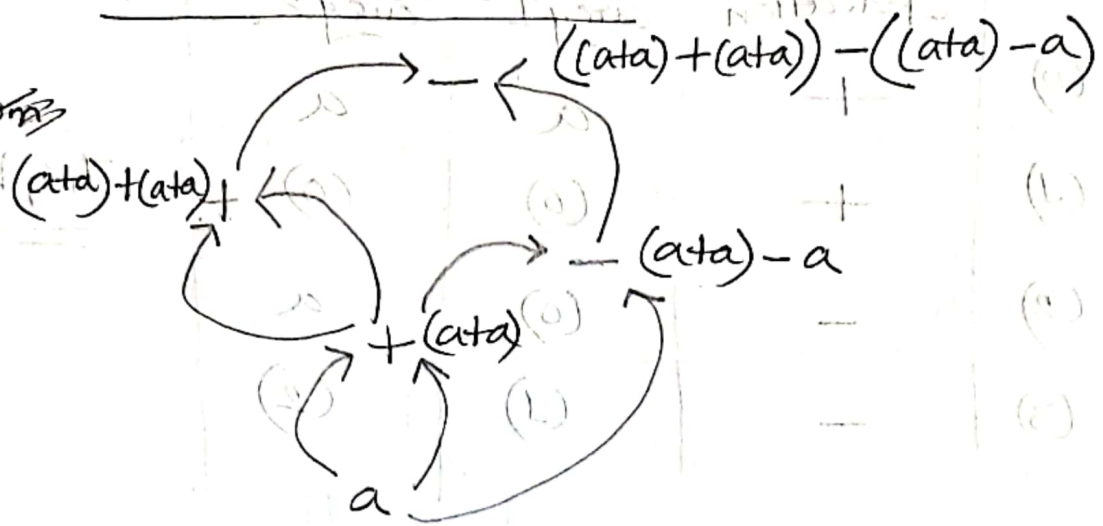
Ans to the Q. No. 2



08

Ans to the No. 3

(a) From



∴ The total expression:

$$((ata)+(ata)) - ((ata)-a)$$

3 address code for the DAG:

(0) $t_1 = a + a$

(1) $t_2 = t_1 + t_1$

(2) $t_3 = t_1 - a$

(3) $t_4 = t_2 - t_3$

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(b)

steps	operation	arg1	arg2	result
(0)	+	a	a	t_1
(1)	+	t_1	t_1	t_2
(2)	-	t_1	a	t_3
(3)	-	t_2	t_3	t_4

Quadruples

	Operation	arg1	arg2
(0)	+	a	a
(1)	+	(0)	(0)
(2)	-	(0)	a
(3)	-	(1)	(2)

Triples

✓ : notice arg1, arg2, result
 $(a - (a + a)) - ((a + a) + (a + a))$

: DNF: all - not - good example

$$a + a = t_1$$

$$t_1 + t_1 = t_2$$

$$a - t_1 = t_3$$

$$t_3 - t_2 = t_4$$

Quadruples

result	opnd1	opnd2	operation	code
t1	a	a	+	(0)
t2	t1	t1	+	(1)
t3	a	t1	-	(2)
t4	t3	t2	-	(3)